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DUNMAN HIGH SCHOOL **Preliminary Examination** **Year 6**

H2 BIOLOGY

9648/03

Paper 3 Applications Paper and SPA Planning Task

25 September 2014

2 hours

Additional Materials: Writing paper

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section A:

Consists of 4 Structured Questions

Answer **all** questions.

Write your answers in the **space provided** on the question paper.

Section B:

Consists of 1 SPA Planning Task

Write your answers on the separate **writing papers** provided. At the end of the examination, fasten all your work securely together.

Section C:

Consists of 1 Free-Response Question.

Write your answers on the separate **writing papers** provided. At the end of the examination, fasten all your work securely together.

Sections A, B and C are to be submitted **separately**.

INFORMATION FOR CANDIDATES

Essential working must be shown.

The intended marks for questions or parts of questions are given in brackets [].

For Examiner's Use	
Section A [40]	
1	/ 12
2	/ 10
3	/ 12
4	/ 6
Section B[12]	
Section C [20]	
Total [72]	

This document consists of **15** printed pages and **1** blank page.

Section A: Structured Questions (40 marks)

Answer **all** questions in this section.

For
Examiner's
use

Question 1

- (a) You are studying a specific gene named LC1 in mice, and you want to express that mouse gene in *E. coli*. To do that you must insert the mouse gene into the bacterial plasmid using restriction enzymes.

These are the restriction enzymes you can use. Their respective restriction sites are indicated as X: for XbaI, N: for NdeI, S: for Sall, E: for EcoRI and K: for KpnI.

Fig 1.1 is a map of the area of the mouse genome surrounding LC1 in which you are interested.

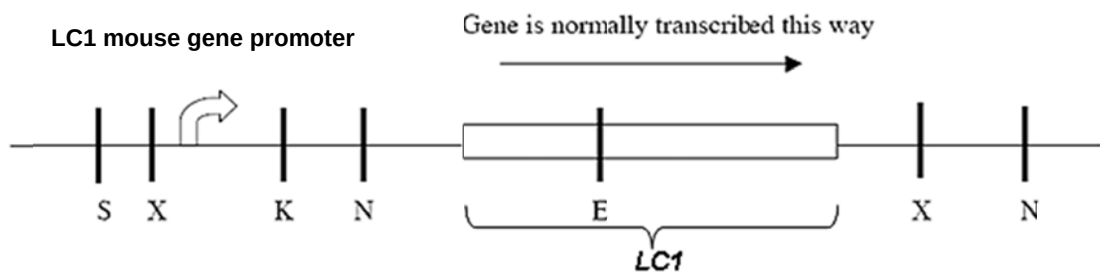


Fig. 1.1: Map of LC1 mouse gene and adjacent regions. Curved arrow indicates the position of the LC1 mouse gene promoter.

Below is the map of the cloning vector which has an ampicillin resistance gene somewhere distant from the restriction sites.

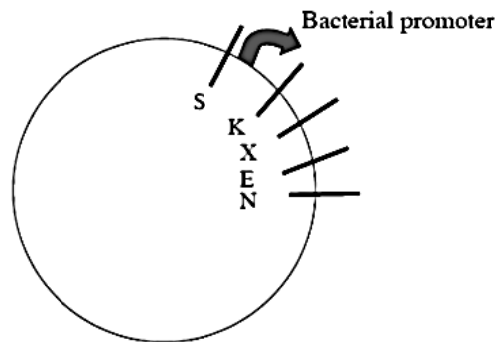


Fig 1.2: Map of cloning vector used, ampicillin resistance gene is present in the vector but not shown

- (i) Besides the regions in the cloning vector already mentioned, state one other region in the vector that is required and why it is necessary. [2]

- (ii) After much consideration, NdeI was used to cut both the gene of interest and the vector. Explain why NdeI was chosen to cut the gene of interest and the vector. [2]

For
Examiner's
use

- (iii) Three possible ligated DNA products that may result from the ligation which will confer ampicillin resistance to the transformed bacteria are shown below.

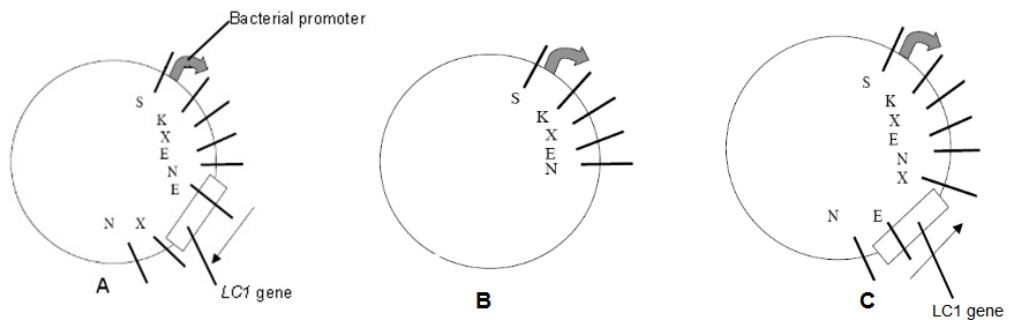


Fig 1.3

Which of the three ligated DNA products is the most likely to allow for successful production of the mouse protein from the bacterial cells? Explain. [3]

- (b) Variable Number Tandem Repeats (VNTRs) and Short Tandem Repeats (STRs) are specific classes of human DNA polymorphisms that can be used for DNA fingerprinting. To obtain the DNA fingerprint of an individual, three VNTR loci **1** to **3** located on chromosome 13, as shown in **Fig. 1.4**, were examined by Restriction Fragment Length Polymorphism (RFLP) analysis. The arrows mark the cleavage sites of *EcoRI*.

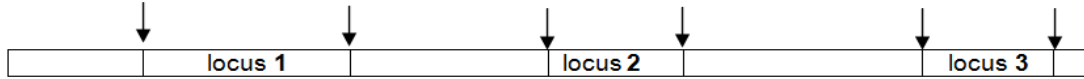


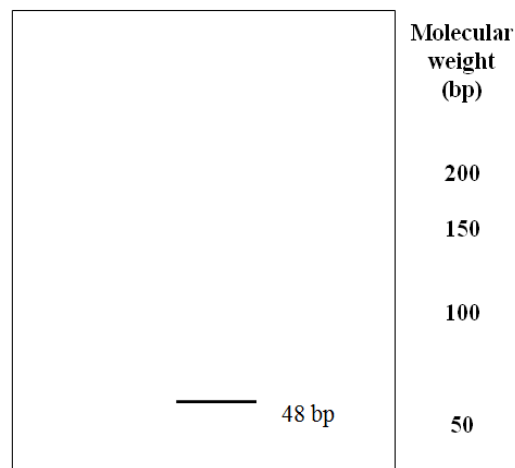
Fig. 1.4

Each VNTR locus contains a range of alleles defined by different numbers (**n**) of tandem repeats of a particular core sequence. The repeat sequences and their respective lengths (in bp) as well as values of **n** for the individual's VNTR loci **1** to **3** are shown in **Table 1**.

Table 1

VNTR locus	Core sequence	Length (bp)	n
1	5'-TAACCGATCGGCAATG-3'	16	3 and 7
2	5'-CCGATCGGCATCAGC-3'	15	4 and 12
3	5'-ACCCGATCGGCA-3'	12	10 and 15

- (i) In the human population, **n** may range between 3 to 20 for VNTR locus **1**. Give the expected number of alleles of VNTR locus **1** that would occur in the human population. [1]
-
- (ii) DNA fingerprinting by RFLP analysis is based on the Southern blotting technique. Write the sequence of a 10-bp probe molecule that can be used to detect all three VNTR loci **1** to **3**. [1]
-
- (iii) Draw the expected results on a Southern blot below, indicating the molecular weight of each expected band. One of the expected bands is shown. [2]



- (iv) If three other VNTR loci with n values not more than 5 are used, suggest with a reason the effect on the discriminating power (i.e. ability to distinguish between individuals) of RFLP analysis using these loci. [1]

*For
Examiner's
use*

Total: [12]

Question 2

- (a) Fig 2.1 below shows the isolation of a specific human gene and cloning into a bacteriophage.

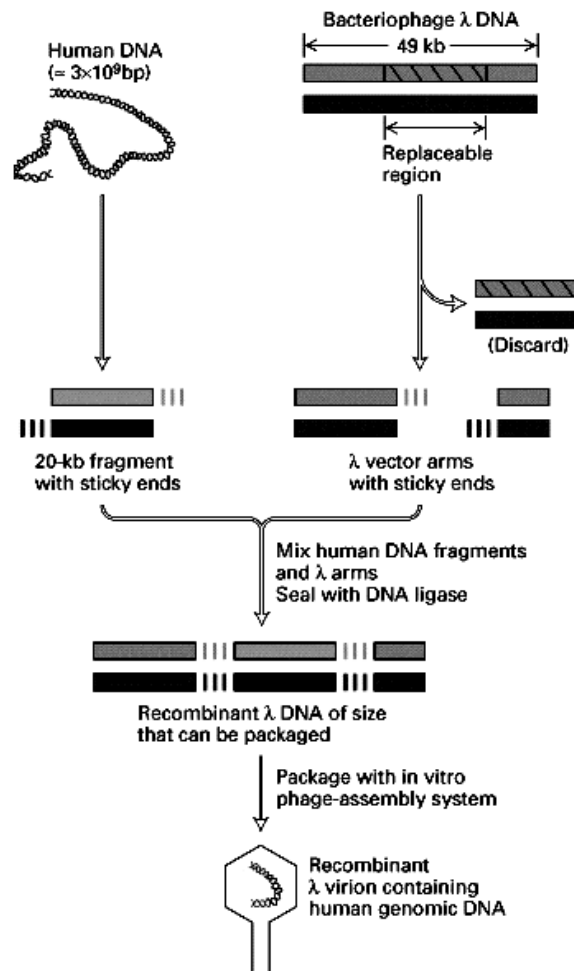


Fig 2.1

- (i) Identify the type of DNA library derived from this. [1]

- (ii) Name another vector that can be used for the making of DNA library and explain why this vector might be favoured over a bacteriophage vector. [2]

(iii) Distinguish between a genomic and cDNA library. [3]

(b) Polymerase Chain Reaction (PCR) is a relatively simple and inexpensive tool that you can use to copy a segment of DNA for many times. PCR is used every day to diagnose diseases, identify bacteria and viruses, match criminals to crime scenes, and in many other ways. **Fig 2.2** below shows the temperature-based graph of 2 PCR thermal cycles.

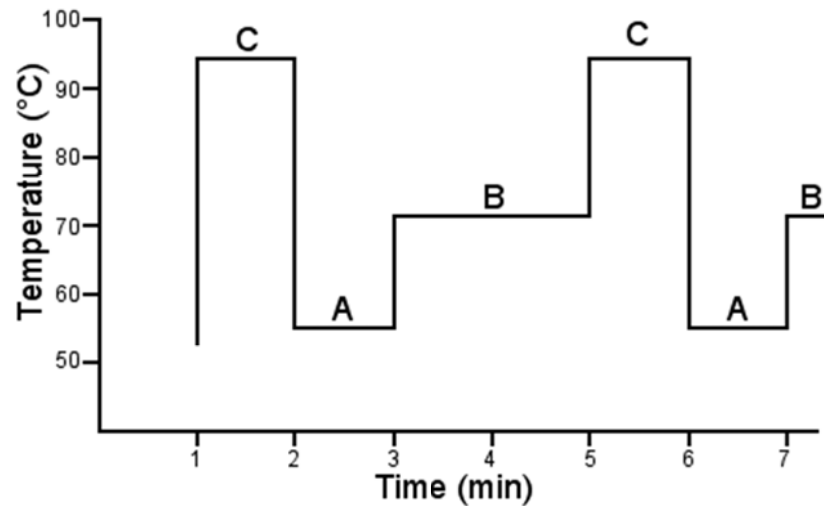


Fig 2.2

(i) With reference to **Fig 2.2**, describe what is happening at stages A and B. [2]

(ii) Suggest why the optimum annealing temperature changes with different DNA target sequences. [1]

(iii) Primers added into the mixture are usually in excess. Explain why. [1]

Total: [10]

Question 3*For
Examiner's
use*

- (a)** Osteogenesis imperfecta (brittle bone disorder), is a genetic disorder characterised by bones that break easily.

Adult bone stem cells transplanted into children with osteogenesis imperfecta have stimulated growth of bone in these children. During the 6 months immediately following the transplant, the children's growth reached 60% to 94% of expected normal values for children their age.

- (i)** In terms of differentiation potential, explain how an adult bone stem cell is different from a fertilized egg of a human. [2]

- (ii)** Explain how the adult bone stem cells may increase bone growth in children with osteogenesis imperfecta. [2]

- (b) Plant physiologists attempted to produce papaya plants using tissue culture.

They investigated the effects of different concentrations of two plant growth factors on small pieces of the stem tip from a papaya plant. Their results are shown in **Table 3**.

For
Examiner's
use

Table 3

Concentration of auxin / $\mu\text{mol dm}^{-3}$	Concentration of cytokinin / $\mu\text{mol dm}^{-3}$		
	5	25	50
0	No effect	No effect	Leaves produced
1	No effect	Leaves produced	Leaves produced
5	No effect	Leaves produced	Leaves and some plantlets produced
10	Callus produced	Leaves and some plantlets produced	Plantlets produced
15	Callus produced	Callus and some leaves produced	Callus and some leaves produced

- (i) With reference to **Table 3**, give evidence that the cells from the stem tips are totipotent. [2]

- (ii) Stem tips are used in tissue culture to produce disease-free plants.

High auxin levels are known to be able to inhibit viral infection in plant cells.

Fig 3 below shows the concentration of auxin in different cells of plants.

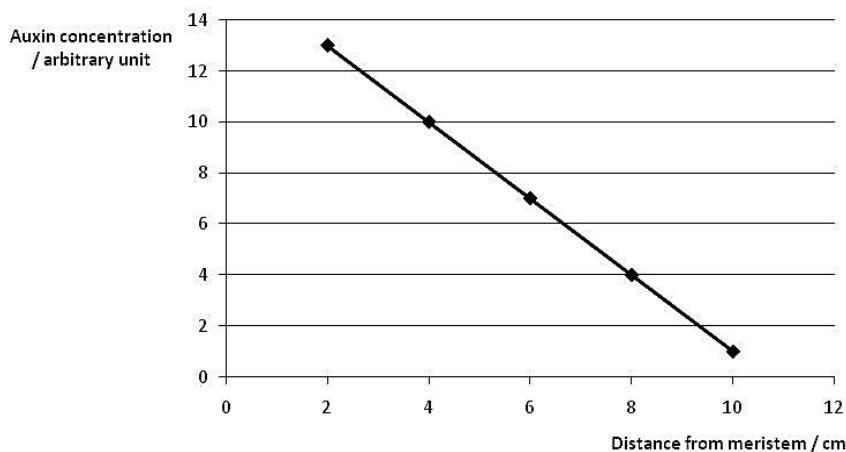


Fig 3

With reference to **Fig 3**, suggest why cells are taken from stem tip instead of flowers or leaves for tissue culture. [2]

- (iii) State the recommended cytokinin to auxin ratio that should be used to produce papaya plants using stem tip culture. [1]

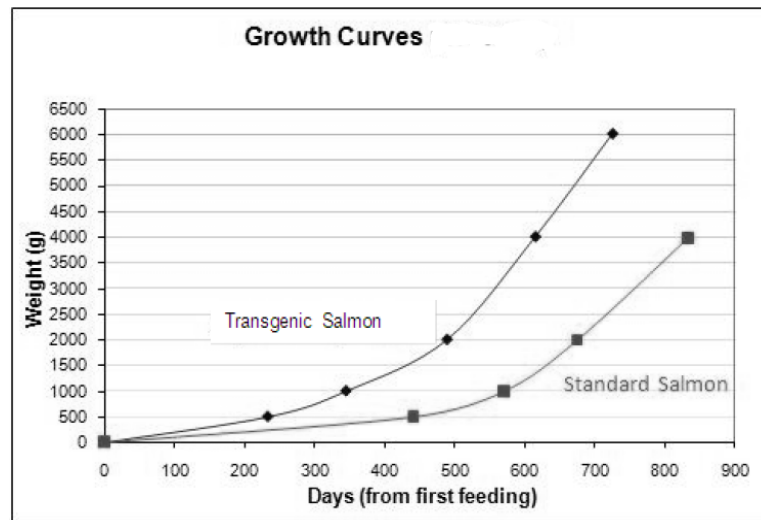
- (iv) Explain the advantages of growing papaya plants from tissue culture rather than from seeds. [3]

Total: [12]

Question 4

- (a) In 1989, a transgenic salmon line was created by injecting eggs of Atlantic salmon with a gene construct containing the gene of the Chinook salmon growth hormone. Transgenic salmon were grown as sterile, all-female population in tanks on land and located away from the sea.

Salmon reaches a marketable value at 4kg. The growth of transgenic salmon and the standard Atlantic salmon is shown in **Fig 4**.

**Fig4**

- (i) With reference to **Fig 4**, explain the benefits of farming transgenic salmon as compared to the standard Atlantic salmon. [3]

- (ii) Give reasons why transgenic salmon are grown as sterile, all-female populations in tanks on land away from the sea. [3]

Total: [6]

Section B: SPA Planning Task (12 marks)

Write your answers on the writing paper provided.

A **NIL RETURN** is required.

Question 5

Penicillin is an antibiotic that prevents the synthesis of mucopeptides (crystal lattice structure made up of linear chains of alternating amino sugars) in cell walls made of peptidoglycan by preventing the formation of peptide bonds. It is bactericidal (kills bacteria) and is effective against gram-positive bacteria.

Staphylococcus epidermidis is a gram-positive bacterium which forms small white colonies. Discs containing penicillin can be placed on an agar plate containing *Staphylococcus epidermidis*. If the penicillin has been effective there will be a clear zone around the disc as shown in **Fig 5**.

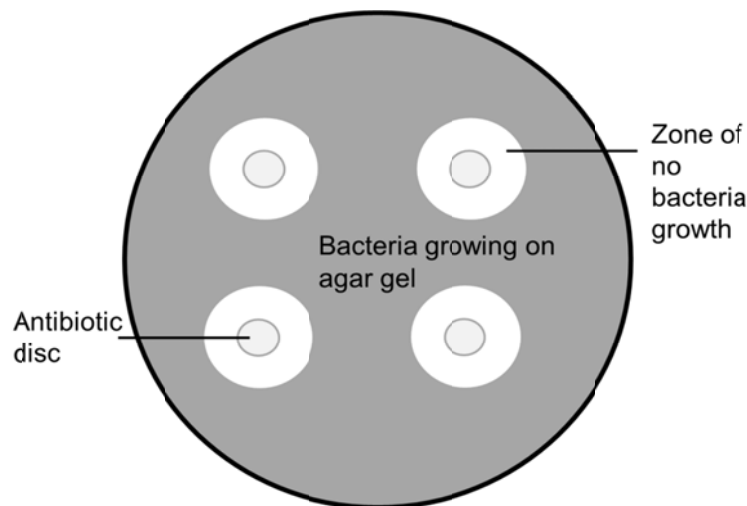


Fig 5

Plan an experiment to investigate the effectiveness of different concentrations of penicillin on the growth of *Staphylococcus epidermidis*.

Your planning must be based on the assumption that you have been provided with the following equipment and materials:

- Bunsen burner, to enable good aseptic conditions
- Bacterial culture in nutrient broth
- Molten nutrient agar
- Sterile water
- Sterile 90mm Petri dishes
- Sterile loops
- 1cm³ pipette
- Filter paper discs
- Forceps
- 1% penicillin solution
- Bactericidal disinfectant for containment of used forceps and pipettes, also to clean work surfaces

Your plan should include:

- a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- an explanation of theory to support your practical procedure
- an explanation of the dependent and independent variables involved
- relevant, clearly labeled diagrams, if necessary
- proposed layout of results tables and graphs with clear headings and labels
- correct use of scientific and technical terms
- safety measures to minimise any risks associated with the proposed experiment

Total: [12]

Section C: Free-Response Question (20 marks)

Write your answers on the writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL RETURN** is required.

Question 6

- (a)** With reference to genetic diseases, describe SCID (severe combined immunodeficiency). [6]
- (b)** Describe the treatment of SCID and cystic fibrosis, using the relevant viral and non-viral gene delivery systems. [8]
- (c)** Discuss the social and ethical considerations for the use of gene therapy. [6]

Total: [20]

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**DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION 2014
YEAR SIX
H2 BIOLOGY (9648)
PAPER 3**

Suggested Answers

Structured Answers

1(a)(i)

Origin of replication

Sequence to which DNA polymerase (and helicases) will bind to initiate replication of the vector with the gene insert / recombinant vector OR so that plasmid can replicate independently

OR

Another (or named – not Ampicillin) antibiotic resistance gene / β -galactosidase gene within the region

where the restriction sites are so that the insertion of LC1 at the restriction site will disrupt this antibiotic resistance gene / β -galactosidase gene and allow for selection of bacteria with the recombinant plasmid

(ii)

There are two NdeI restriction sites flanking / each on either side of the gene of interest and so will cut out the whole gene of interest from the surrounding genome

Also cutting at the two NdeI restriction sites will exclude the mouse promoter

good because the bacterial promoter is the one that is required to initiate transcription in the bacterial cell / mouse promoter not needed to initiate transcription, cannot initiate transcription

(iii)

A;

You want the bacterial promoter to drive transcription of the mouse gene in the correct orientation so that the correct strand of DNA is used as a template in transcription

B is the vector re-ligated to itself and does not have the gene inserted

C would result in the wrong strand of DNA being used as the template for transcription

(b)(i)

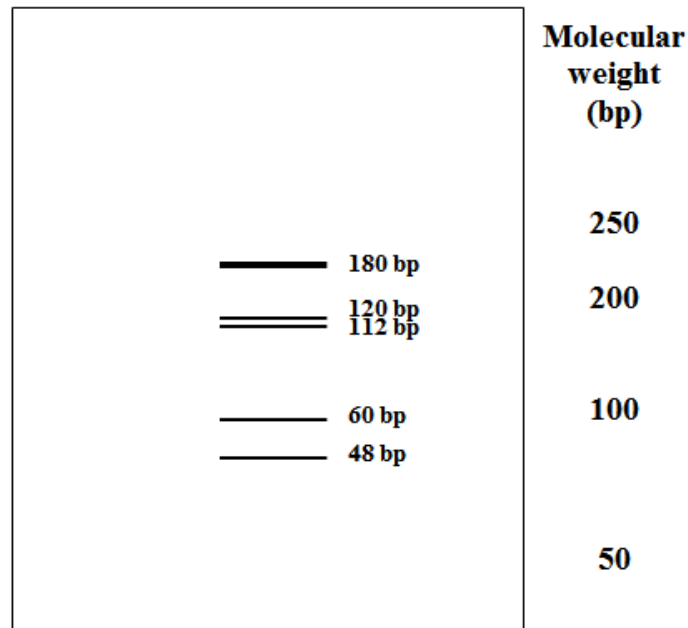
18

(ii)

3'- GGCTAGCCGT – 5';; OR 5'-TGCCGATCGG-3';; (DNA probe) OR

3'- GGCUAGCCGU – 5';; OR 5'-UGCCGAUCGG-3';; (RNA probe)

(iii)



Correct positions of four bands, corresponding to 60 bp, 112 bp, 120 bp and 180 bp, with respective molecular weights indicated with 180-kb band thicker than the rest of the bands

(iv)

Discriminating power would be lowered / reduced as with a smaller n value / number of alleles or allelic variants or polymorphisms at each locus, the number of allelic combinations at each locus would be smaller

2(a)(i)

genomic library

(ii)

Plasmid contain an origin of replication (ori) and therefore foreign DNA fragment will be replicated together

can be engineered to contain genetic markers which confer well-defined phenotypes on the host cell for selection

plasmid do not require packaging process like the bacteriophage vector;

able to accept a larger insert

(iii)

	Genomic library	cDNA library
Function	It contains the entire genome of an organism including the coding and non-coding sequence.	It contains only the protein coding DNA.
Genetic material isolated	Chromosomal DNA from any cell/tissue.	Total mRNA from a particular cell/tissue type where that particular protein of interest is produced in large quantity.
Key enzyme involved	Genomic DNA is cleaved with restriction enzyme, generating random overlapping DNA fragment.	Reverse transcriptase catalyses reverse transcription of mRNA to a single strand complementary cDNA. Then RNAase is used to hydrolysed the mRNA and DNA polymerase is added to synthesis a complementary DNA strand.
Size of library	There are larger numbers of different types of clones in the library. (fragments originate from whole genome)	There are smaller numbers of different types of clones in the library (approx. 1% of the entire genome)
Intact genes cloned	A gene may contain restriction sites within itself. Because restriction enzymes do not recognize gene boundaries, some genes may be cut and divided up among two or more fragments.	mRNA consists of exons and hence using mRNA as a template to obtain cDNA by reverse transcription, the cDNA obtained is intact genes.
Study sequence	It is used to study the introns or the regulatory sequences associated with a gene.	It is used to study the exact coding sequence of the gene.
Screening of gene	It can be used for screening and isolation of a gene when the cell type in which it is expressed is unknown.	It is used for screening and isolation of a gene that is expressed actively as a functional product in a particular known cell/tissue type.
Types of clone present	There is equal quantity of all DNA fragments.	mRNA type which is more actively synthesized will be at a larger quantity than the other mRNA. Thus, there will be unequal proportion of different cDNA fragments.

(b)(i)

Stage A:

Temperature decreased to 55°C which allows the DNA primers to bind by hydrogen bonding to to anneal to complementary sequences on the template strands

Stage B:

The mixture is heated to 72°C where Taq polymerase will extend the primers by adding deoxyribonucleotides to the 3' OH ends of the primers

(ii)

The temperature is largely determined by the A+T and G+C composition of primers where primers with high content of G + C base pairs can hybridize at higher temperatures than primers with a high A + T content.

(iii)

This is to ensure that primers anneal to the separate DNA strands before the two DNA strands rewind back together

3(a)(i)

The zygote/fertilized egg is totipotent i.e. it can give rise to all cells types in the body including extra embryonic tissue

but the adult bone stem cell is multipotent i.e. it can only give rise to a limited range of cell types

(ii)

Stem cells can divide by mitosis

and in the presence of the appropriate signals can also differentiate into new bone cells; This thus increases the number and type of bone cells present which contribute towards bone growth

(b)(i)

Quote any values that produce plantlets, For example, at 10 $\mu\text{mol dm}^{-3}$ auxin and 50 $\mu\text{mol dm}^{-3}$ cytokinin, whole plantlets (with leaves and roots) are produced from the stem tips

The stem tip cells are totipotent and hence able to differentiate into all specialized cell types to form a whole plant

(ii)

Quote: As distance increases from 2 to 10 cm, auxin concentration decreases from 13 to 1AU, auxin concentration is highest nearest the stem tips because auxin acts as plant growth hormone present in meristemic cells

Auxin has dual function and also acts an inhibitor / repressor of viral replication

The vascular bundles which usually transmit diseases are not yet established at stem tips but are present in leaves and flowers

(iii)

5 : 1 (with 50 $\mu\text{mol dm}^{-3}$ cytokinin and 10 $\mu\text{mol dm}^{-3}$ auxin)

(iv)

Growing papaya from tissue culture produces clones which are genetically identical to the parent

Allow mass propagation of clones

Hence all the favourable traits/desired characteristics of the parent are found in these clones

Whereas in seeds these favourable traits of the parent may not be found in the progeny due to genetic variation from independent assortment/ crossing over at meiosis/ random fusion of gametes at fertilization

(AVP) Tissue culture allows genetic modification/engineering to form favourable trait/phenotype

4(a)(i)

The transgenic salmon reaches marketable value at 4kg much earlier at 600 days from first feeding compared to the standard salmon which reached marketable value at 830 days from first feeding / Ref to higher rate of growth (quote data) / Ref to higher mass within a certain time period (quote data)

The faster growth allows farmers to save on cost of feeding the fish/ higher feed conversion efficiency

The also allows a higher turnover rate of fish farmed thus increasing the supply/yield of salmon meat in a certain period

(ii)

Prevent transgenic salmon from escaping into the sea/river in a land based facility;

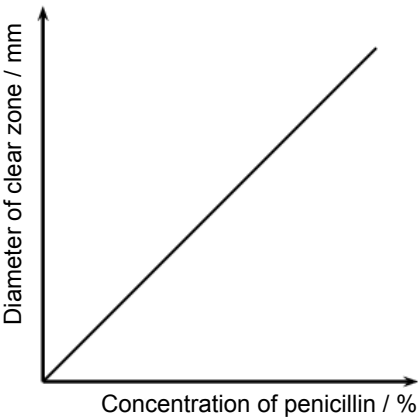
Prevent reproduction of transgenic salmon

Or

Prevent transgenic salmon crossing with standard salmon pass on the transgenes to wild salmon

Prevent it from outcompeting the standard salmon for resource and mates

Planning Answer

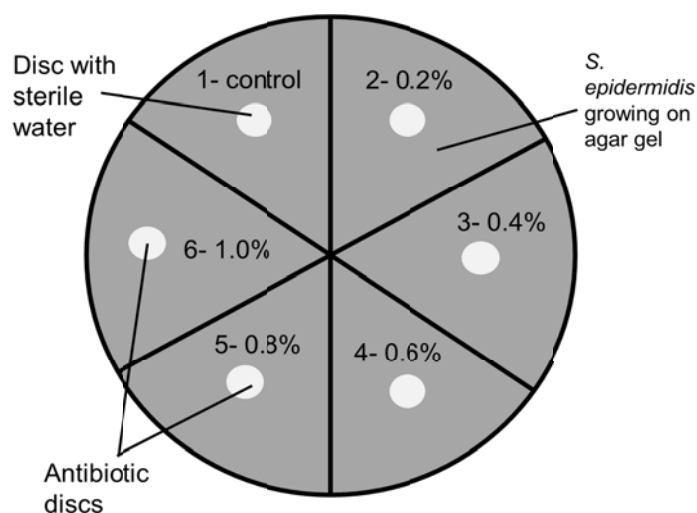
Theory	Penicillin is an antibiotic that interferes with the synthesis of peptidoglycan in bacterial cell wall by breaking down the formation of peptide bonds. This greatly <u>weakens the cell wall</u> and causes the <u>bacterium to lyse or burst open</u>, because of osmotic pressure. As a result, this directly <u>kills the bacteria</u>. In this experiment, penicillin is placed in the form of antibiotic discs on the agar gel plated with <i>Staphylococcus epidermis</i>. The <u>size of the clear zone</u> formed after incubation is a measure of the <u>effectiveness of the penicillin</u>. Increasing concentration of penicillin will increase the diameter of the clearing. The expected trend is as such.  The <u>increase in the diameter of the clear zones should be directly proportional to the concentration of penicillin</u>.	✓ description of scientific reasoning and theory of the method used to measure effectiveness of penicillin ✓ expected relationship between [penicillin] and diameter of clear zone
Variables	Independent variables: <u>5 concentrations of penicillin solution</u> (0.2, 0.4, 0.6, 0.8, 1.0 %) prepared by simple dilution. Dependent variables: <u>Diameter of clear zone / mm</u> Controlled variables (<u>any 2</u>): • Concentration and volume of bacterial culture • Concentration and volume of agar used • Size of filter paper disc • Weight of filter paper	✓ independent, dependent variables and controlled variables

Procedure

1. Add 10cm³ of nutrient agar to the petri dish using aseptic technique and leave to set.
2. Prepare penicillin solutions of 5 different concentrations using simple dilution according to the table below. Label the beakers accordingly.

Concentration of penicillin solution / %	Volume of 1% penicillin stock solution / cm ³	Volume of sterile or distilled water / cm ³	Total volume / cm ³
1.0	10	0	10
0.8	8.0	2.0	10
0.6	6.0	4.0	10
0.4	4.0	6.0	10
0.2	2.0	8.0	10

3. Draw lines on the base of the petri dish so that the base is split into 6 equal parts. Label the sections 1 to 6.
4. When the gel is cast on the petri dish, use a sterile loop to plate the nutrient broth containing *Staphylococcus epidermidis* onto the agar gel.
5. Prepare a control disc by soaking a sterile disc in sterile water.
6. Using flamed and cooled forceps, place the control disc in the centre of section 1 using flamed and cooled forceps.
7. Prepare the 5 penicillin discs containing 5 different penicillin concentrations by soaking each of the discs into the 5 different concentrations of penicillin solutions prepared from simple dilution.
8. Using flamed and cooled forceps place the disc containing the 5 concentrations into the other 5 labeled sections.
9. The setup of the petri dish in the experiment is as shown below:



✓ dilution method

✓ control

✓ method-setup

✓ relevant diagram

10. Conduct 3 replicates from steps 1-8 and 2 repeats for all prepared concentration of penicillin.
11. Seal the petri dish and incubate the petri dish at 30°C for 2 days.
12. Without opening the lid measure the diameter of the clear zone around each disc using vernier calipers or a ruler.
13. Tabulate the results to show the effect of different penicillin concentrations on *Staphylococcus epidermidis* growth.

✓ 3 replicates, 2 repeats

✓ fixed incubation time

✓ method to calculate growth of bacteria

✓ appropriate table and graph

Table showing the effect of different concentration of penicillin/% on the diameter of clear zone /mm

Conc. of penicillin	Diameter of clear zone /mm			
	Reading 1	Reading 2	Reading 3	Average
1.0				
0.8				
0.6				
0.4				
0.2				
0.0 (Control)				

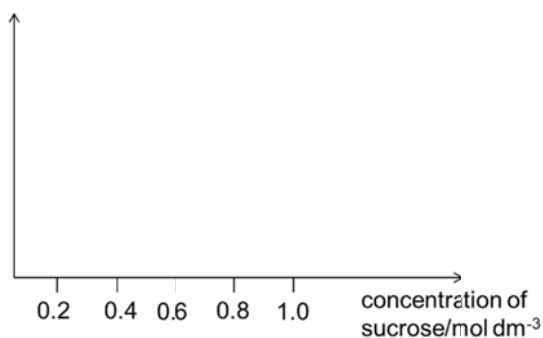
Average = $\frac{\text{reading 1} + \text{reading 2} + \text{reading 3}}{3}$

3

14. Plot a graph of diameter of clear zone /mm produced against the concentration of penicillin /%.

Graph of diameter of clear zone /mm produced against the concentration of penicillin /%.

Diameter of clear zone / mm



Risk and precaution	To prevent infection or growth of harmful microorganisms: • Cover all cut or broken skin with a waterproof dressing • Wear tightly fitting disposable gloves and clean laboratory coat • Clean the bench surface with bactericidal disinfectant and use a bunsen burner which creates a sterile environment. Students should work as close as possible to the flame. • Swap any spillages with bactericidal disinfectant. • Proper disposal / treatment of contaminated materials or equipment using sterilizer/autoclave	✓ Safety (Any 1)
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Essay Answers

6 (a) With reference to genetic diseases, describe SCID (severe combined immunodeficiency). [6]

Marking Point	Answer
1	Sex-linked recessive SCID;
2	Mutation in interleukin receptor gene on the X chromosome;
3	Mutation results in a non-functional common gamma chain;
4	Non-functional receptor unable to bind ligand interleukins;
5	Lymphocytes fail to develop or are non-functional;
6	Autosomal recessive SCID;
7	Mutation in adenosine deaminase (ADA) gene on chromosome 20;
8	Mutation results in a defective ADA;
9	Defective ADA unable to break down toxic deoxyadenosine, toxic metabolite accumulates and lymphocytes die;
10	Individuals with either forms of SCID have low counts of functional lymphocytes and are immunocompromised;

6 max

6 (b) Describe the treatment of SCID and cystic fibrosis, using the relevant viral and non-viral gene delivery systems. [8]

Treatment of SCID

Marking Point	Answer
11	An ex vivo gene therapy approach can be used to treat both ADA-SCID and sex-link-SCID;
12	remove T-lymphocytes from patient with SCID and insert healthy genes in them using a <u>retrovirus</u> , then transplanting them back into the patient;
13	where lymphocytes obtained from the blood of the patient / bone marrow stem cell and stimulated to proliferate in culture;;
14	A functional adenosine deaminase (ADA) cDNA / IL2-RG cDNA in a retroviral vector was then introduced into these cells;
15	The retrovirus had been inactivated to become non-pathogenic;
16	thus the retrovirus was integrating the normal ADA gene / IL2-RG gene into the chromosomes of the patient's cells;
17	The transgenic lymphocytes were allowed to proliferate further in the culture and reintroduced into the patient's body;

4 max

Treatment of CF:

Marking Point	Answer
18	An in vivo gene therapy approach can be used to treat cystic fibrosis;
19	Introduce the normal CFTR gene into liposomes ;
20	Spray these liposomes into the nose and mouth of CF patients as an aerosol or administering liposomes into the airways <i>via</i> a hose;
21	Liposomes fuse with lipids in cell membranes of tracheal cells releasing the normal CF gene into the cytoplasm of cells;
22	The normal CFTR gene is transcribed into mRNA, and the normal protein translated;
23	The normal CFTR protein enters cell membranes and starts to transport chloride ions out of cells, thereby thinning the mucus and alleviating CF symptoms;
24	As the gene expression is transient, the drug must be re-administrated often for the treatment to be effective;

4 max

6 (c) Discuss the social and ethical considerations for gene therapy. [6]

Marking Point	Answer
25	Potential for treating ill patients or preventing onset of severe illness (when conventional therapy has failed, gene therapy is the only hope for patients . Compared to the hardship and death faced by these patients, the risks and uncertainties of gene therapy are acceptable; civic, religious, scientific and medical groups have agreed that somatic-cell gene therapy is appropriate for humans).
26	Somatic-cell gene therapy corrects the genetic defect present in the individual alone and does not impact genetic information passed on to the next generation.
27	Somatic-cell gene therapy is like any other new medical technology .
28	To prevent research on gene therapy is to <u>infringe</u> upon intellectual freedom of researchers .
29	With current rate of progress in gene therapy techniques risks become more identifiable and outcomes can be predicted rather precisely .
30	(Specifically referring to <i>germ-line gene therapy</i>) It offers true cure , not just a relief of symptoms or a palliative treatment.
31	It may be the only effective way of addressing some genetic diseases by <u>preventing transmission of diseased genes</u> .
32	The expense of somatic-cell therapy for multiple generations is avoided.
33	Medicine should respond to the reproductive health needs of prospective parents at risk for transmitting serious genetic diseases.
34	The scientific community has a right to free inquiry within the bounds of acceptable human research.

35	Once successful techniques are developed, gene therapy could help parents and researchers avoid the moral dilemma of disposing “defective” embryos in the laboratory by “repairing” them instead.
36	It is difficult to distinguish between “good” and “bad” uses of gene modification techniques.
37	It is important to consider between genetic disease and genetic attribute . Consideration should be given beyond the therapeutic if there is only a genetic tendency towards a particular disease.
38	There is potential for harmful abuse of the technology. It should be a consensus that gene therapy is therapy , i.e. the correction of <i>bona fide</i> disease conditions, rather than enhancement , which would mean “improving the human species”. Some also fear that widespread use of this technology could lead to new definitions of “normal” that would exclude individuals who are, for example, of merely average intelligence. And, justly or not, some people associate all genetic manipulation with past abuses of the concept of “eugenics”, or the study of methods of improving genetic qualities through selective breeding.
39	There are many unresolved questions raised by gene therapy: • Can gene expression be controlled? • What happens if therapeutic genes are overexpressed or if a gene shuts off shortly after it has been introduced? • How long will the therapy last? • Whether a treatment will require frequent administration of the therapeutic gene? If a disease can be treated by only correcting a small number of body cells or a majority of diseased cells must be corrected?
40	There is difficulty in following patients in long-term clinical research , including surveillance of future generations to monitor the long-term effects of gene therapy.
41	Many gene therapy candidates are children who are too young to understand ramifications of gene therapy
42	There are potential conflicts of interest of individuals who wish to protect their privacy and confidentiality versus interests of insurance companies or society in not bearing the financial burdens of caring for children with serious genetic defects.
43	There are issues of justice and allocation : can we afford the expense of gene therapy? Who should receive gene therapy? If only those who can afford it, where is the principle of justice? The procedures now being tested are expensive and require expertise and equipment found only in major medical centers. A related question is, “Should gene therapy be reserved for treating serious diseases?” “Who should pay for their use?” “Should we skew the distribution of desirable biological traits?”
44	(Specifically referring to germ-line gene therapy) ▪ Long-term effects of germ-line therapy are not known. ▪ It opens the door to attempts at altering human traits not associated with disease (i.e., eugenics), which in turn leads to issues of increased social inequality and a lowered tolerance for genetic diversity. ▪ Violates rights of future generations to inherit a genetic

	endowment which has not been intentionally modified. ▪ Too expensive. Essentially mates generations of unconsenting research subjects (i.e., germ-line therapy involves research on early embryos and affects their offspring).
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6 max